

ADARSH PRATAP SINGH

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Professional Summary : Data Scientist and Machine Learning Engineer with experience in **building predictive models**, ETL pipelines, Data Structure & Algorithms and full-stack applications. Proficient in Python, **TensorFlow**, **PyTorch**, **OpenCV**, **RAG**, SQL, **LLM** and **scikit-learn**. Skilled in REST APIs, **Flask**, **Django**, and web development (HTML, CSS, JavaScript). Hands-on with cloud platforms (**AWS**, **GCP**), Docker, MLOps, and **CI/CD**. Strong background in data visualization, statistics, and business intelligence.

Experience

IBM | Data Science Internship

Lucknow | Oct 2024 – Dec 2024

- Performed ETL processes on large, structured and unstructured datasets for **data cleaning, transformation, and preprocessing**, enabling efficient data analysis and machine learning workflows.
- Applied machine learning algorithms, including linear regression, **logistic regression**, **classification**, **clustering**, and ensemble methods (e.g., **Random Forest**, **XGBoost**) to build accurate predictive models for business insights.
- Engineered features, created feature selection and **dimensionality reduction pipelines** (e.g., PCA), and optimized data pipelines using Python, **Pandas**, **NumPy**, and **SQL** to improve model performance, scalability, and deployment readiness.

Projects

Plant Identification System (Developed a CNN-based tool to identify plants from images)

Tools | **OpenCV**, **TensorFlow/ Keras**, **FastAPI**, **PostgreSQL**, **MobileNetV2**, **Deep Learning**

- Built **CNN-based** plant species classifier with **robust preprocessing and augmentation**.
- Designed RESTful APIs for **real-time predictions** and **frontend integration**.
- Integrated PostgreSQL** to manage species taxonomy data.

Student Performance Prediction System (Built a machine learning model to predict student scores using regression techniques)

Tools | **Python**, **Machine Learning**, **NumPy**, **Pandas**, **Scikit-learn**

- Built a predictive model to estimate student performance using academic and behavioral data.
- Performed data preprocessing, including handling missing values, **encoding categorical variables**, and **feature scaling**.
- Implemented and trained **regression algorithms** (e.g., Linear Regression) for score prediction.
- Evaluated model performance using metrics like **Mean Absolute Error (MAE)** and **R² score**.
- Applied basic model optimization techniques to improve prediction accuracy.

Sports Match Report Generator(AI tool with Flask & RAG to turn PDF match scorecards into auto-generated news articles)

Tools | **PyMuPDF**, **Hugging Face Transformers**, **SentenceTransformers**, **FAISS**, **RAG Pipeline**, **Flask**, **Gradle**, **PDF Parsing**, **LLM Orchestration**, **Vector Store**, **NLP**, **Chatbot Development**

- Developed an AI-driven Sports Match Report Generator using Flask, RAG pipeline, FAISS vector store, and LLM text generation to automatically **create news-style articles** from PDF match scorecards.
- Implemented **PDF parsing**, **text chunking**, **embedding vectorization**, **semantic search**, and a Flask-based **chatbot** for dynamic match summaries and star player insights.
- Demonstrated end-to-end orchestration of NLP pipelines, combining information retrieval, **LLM prompting**, and automated content generation for scalable sports media workflows.

Certifications

- A-Z Machine Learning (Udemy)
- Python 101 for Data Science (IBM)
- Java Programming: Beginner to Advanced (Udemy)
- IBM Cloud Essentials (IBM)
- IBM Big Data Foundations – Level 1

Skills

- Languages**: Python, Java, HTML, Matlab
- Data Visualization**: Seaborn, Matplotlib, PowerBI
- Databases**: PostgreSQL, MongoDB, MySQL, Apache Spark
- Cloud & Tools**: AWS (S3, SQS, SNS), GitHub Actions, CI/CD, Git
- Data Science & ML**: NumPy, Pandas, Scikit-learn, TensorFlow, Keras, OpenCV, ETL pipelines, Regression, Classification, Clustering, Caffe, PyTorch, MXNet

Education

- APJ Abdul Kalam Technical University, Lucknow** | Bachelor of Technology in Computer Science - AI & ML | **64 %**
- Sri Raghukul Vidyapeeth, Gonda** | Senior Secondary School (CBSE) | **64 %**